

Machine Learning — Economics Track

Sample Exam — 25 Practice Questions

Part 1 — Is the concern valid? (Questions 1–10)

1. A member of your team raises the following concern about the pipeline:

"Two of the columns will simply be empty when a new customer applies for a loan."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Valid — avg_days_late and collections_flag describe what happened after the money was lent, so no applicant can have them on decision day
- B. Not valid — the bank collects these fields from the credit bureau at application
- C. Not valid — empty values are handled automatically by the cleaning step
- D. Valid — but the columns concerned are income and credit_score

2. A member of your team raises the following concern about the pipeline:

"A single customer can occupy several rows of the table, which the split does not take into account."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Not valid — the pipeline removes duplicate customers before splitting
- B. Valid — the table holds monthly snapshots, so most customers appear 3–8 times, and a random row-level split scatters them across train and test
- C. Not valid — repeated rows only slow training down, they do not affect the score
- D. Valid — but this affects only customers who defaulted

3. A member of your team raises the following concern about the pipeline:

"Loans have been observed for different lengths of time, yet they are all labelled by the same rule."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Not valid — the label is measured over a fixed 24-month horizon
- B. Not valid — observation time is irrelevant once the outcome is known
- C. Valid — 'missed 3+ payments at some point' has no fixed window, so a loan observed for three months and one observed for eight are judged on different terms
- D. Valid — but the fix is simply to drop the month column

4. A member of your team raises the following concern about the pipeline:

"There are no examples in the data of the applicants the bank turned down."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Not valid — rejected applications are stored with an estimated outcome
- B. Valid — the warehouse stores approved loans only, so the riskiest slice of the applicant population never appears in training
- C. Not valid — rejected applicants are irrelevant, since we only lend to approved ones
- D. Valid — but this is easily fixed by generating synthetic rejected applicants

5. A member of your team raises the following concern about the pipeline:

"The income column looks as though it holds two different kinds of number."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Not valid — income is a single well-defined quantity by nature

- B. Not valid — the standardisation step resolves any inconsistency in the column
- C. Valid — the column mixes gross and net income, which differ by roughly that ratio
- D. Valid — the histogram shows two populations about twelve times apart, consistent with monthly and annual incomes merged into one column

6. A member of your team raises the following concern about the pipeline:

"The pipeline uses a validation set to decide when to stop training the neural network."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Not valid — there is no validation set at all: the data is split into train and test only, and the network runs a fixed 30 epochs with nothing monitoring it
- B. Valid — and that is the correct use of a validation set
- C. Valid — but the validation set is too small to be reliable
- D. Not valid — the network stops automatically when the loss stops improving

7. A member of your team raises the following concern about the pipeline:

"The Random Forest's results are invalid because its features were standardised before training."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Valid — decision trees cannot be trained on standardised features
- B. Valid — standardisation destroys the thresholds the trees rely on
- C. Not valid — standardising does no harm to a tree-based model; the real problem with that step is that it was fitted on all the data before the split
- D. Not valid — and there is nothing wrong with the standardisation step at all

8. A member of your team raises the following concern about the pipeline:

"The notebook's comments describe the number of trees and the number of epochs as things the model works out for itself."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Not valid — the number of trees is indeed learned during training
- B. Valid — those are hyperparameters the team chose, and calling them 'model parameters' signals that nobody justified the choices
- C. Not valid — the comments describe them correctly as settings
- D. Valid — but only the epochs are mislabelled, not the trees

9. A member of your team raises the following concern about the pipeline:

"The pipeline never checks whether the two models actually agree with each other."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Valid — two models must always be checked for agreement before deployment
- B. Not valid as a priority — agreement between models tells us little; what is missing is far more basic, namely any measure of how each model performs on the customers who default
- C. Valid — disagreement between models is the standard test for data leakage
- D. Not valid — the pipeline does compare their predictions row by row

10. A member of your team raises the following concern about the pipeline:

"The results cannot be trusted because the code was written with the help of an AI assistant."

Reviewing the code and its outputs yourself — is this concern valid?

- A. Valid — AI-written code cannot be used for financial decisions
- B. Valid — AI assistants always introduce data leakage

- C.** Not valid as stated — how the code was written is not itself evidence of anything; the results cannot be trusted because of specific, demonstrable defects in the data and the evaluation
- D.** Not valid — and there are in fact no defects in this pipeline

Part 2 — Approve the right plan (Questions 11–20)

11. The following concern is real and confirmed. The data team proposes four ways forward:

"The same customers appear on both sides of the split, so part of the reported score reflects recognition rather than prediction."

Which plan do you approve?

- A. Increase the test share from 20% to 40%, so the overlap matters proportionally less
- B. Split by customer, so that all of a customer's rows fall on one side only, and treat every score measured before that change as void
- C. Sort the rows by customer before splitting, so that similar rows stay together
- D. Add `customer_id` as a feature, so the models can account for the repetition themselves

12. The following concern is real and confirmed. The data team proposes four ways forward:

"Two features describe events that occur after the loan is granted, and will be empty for every new applicant."

Which plan do you approve?

- A. Keep them and fill them with zeros for new applicants, so the model can still run
- B. Keep them in training but hide them at prediction time
- C. Remove them, rebuild the feature set from information available on submission day, and accept the much lower score that follows as the honest one
- D. Keep them, since removing them causes the accuracy to collapse — which proves how informative they are

13. The following concern is real and confirmed. The data team proposes four ways forward:

"The average used to fill missing incomes was computed over the whole table, before the data was split."

Which plan do you approve?

- A. Compute the filling value from the training rows only, then apply that same value to the test rows
- B. Compute a separate average for each side of the split, so each is internally consistent
- C. Keep the current approach but document it clearly in the notebook
- D. Fill missing incomes with zero instead, which involves no averaging at all

14. The following concern is real and confirmed. The data team proposes four ways forward:

"Missing credit scores were replaced with 0 — a value that cannot occur on a 300–850 scale."

Which plan do you approve?

- A. Replace them with 850 instead, which at least is a value the scale allows
- B. Remove every applicant whose score is missing
- C. Leave the zeros — after standardisation they become ordinary numbers like any other
- D. Fill from a training-set statistic such as the median, and add an indicator column recording that the score was missing

15. The following concern is real and confirmed. The data team proposes four ways forward:

"The label 'missed 3+ payments at some point' has no fixed observation window."

Which plan do you approve?

- A. Keep the definition and weight older loans more heavily, since more is known about them
- B. Fix one horizon for every loan — for example, default within 24 months of origination — document it, and exclude loans not yet observed for that long

- C. Let each modelling experiment choose the window that produces the clearest signal
- D. Add the number of months observed as an extra feature and let the model adjust for it

16. The following concern is real and confirmed. The data team proposes four ways forward:

"The table holds one row per customer-month, while the bank makes one decision per application."

Which plan do you approve?

- A. Keep the monthly rows but train only on customers with at least six months of history
- B. Average each customer's rows into a single row, including the behaviour columns
- C. Rebuild the table with one row per application, containing only what was known on submission day, and retrain both models on it
- D. Keep the monthly rows and predict for every month, then take each customer's worst prediction

17. The following concern is real and confirmed. The data team proposes four ways forward:

"One column carries two different units, depending on which source system produced the record."

Which plan do you approve?

- A. Trace each record back to its source system, convert everything to a single documented unit, and record the definition in a data dictionary
- B. Divide every value above 50,000 by twelve, which will catch nearly all the annual figures
- C. Keep both units — with enough data the model can learn to distinguish them
- D. Drop the income column and rely on the credit score instead

18. The following concern is real and confirmed. The data team proposes four ways forward:

"The team asks to raise the number of trees and the number of training epochs in order to improve the results."

Which plan do you approve?

- A. Approve it, and adopt whichever combination scores highest on the test set
- B. Approve it for the neural network only, since that is the model under consideration
- C. Approve it, and report the average result across all combinations tried
- D. Decline for now: with leaky features and customers on both sides of the split, tuning would only optimise a broken measurement — fix the data first, then tune on a validation set

19. The following concern is real and confirmed. The data team proposes four ways forward:

"The pipeline reports one accuracy figure per model, on data where about 14% of customers default."

Which plan do you approve?

- A. Require the accuracy recomputed after rebalancing the test data to 50/50
- B. Require, per model, a confusion matrix with the share of real defaulters caught and the share of flagged customers who truly default, shown against the accuracy a do-nothing rule would achieve
- C. Require accuracy broken down by month of the loan
- D. Require both test accuracy and training accuracy, so the two can be compared

20. The following concern is real and confirmed. The data team proposes four ways forward:

"The bank wants a model that works for the whole applicant population, not only for profiles the old rules approved."

Which plan do you approve?

- A. Generate synthetic rejected applicants and label them according to their risk scores

- B.** Lower the model's threshold until it begins approving riskier profiles
- C.** Fund a controlled experiment: approve a small random share of borderline applications, track their outcomes, and treat the resulting losses as the cost of acquiring the data
- D.** Retrain the model monthly, so the population corrects itself over time

Part 3 — The model decision (Questions 21–25)

21. The model decision:

"Before any numbers are presented, you must say what would make one model preferable to the other."

Which statement of criteria is defensible?

- A. Overall accuracy on the test set, since it summarises everything in a single number
- B. The share of defaulters caught at a threshold derived from the costs, the money lost per error type, explainability to the regulator, cost to run and maintain, and stability across customer segments
- C. Whichever model represents the more advanced technology, all else being equal
- D. Whichever model the team is more confident maintaining in production

22. The model decision:

"The pipeline's closing line recommends the neural network 'because deep learning is the more advanced technology' — although the Random Forest scored higher in the team's own printout."

What is the significance of this?

- A. It is a minor inconsistency in the write-up and should simply be corrected
- B. It is acceptable: when scores are close, technology maturity is a reasonable tie-breaker
- C. The stated reason is not a criterion, and the recommendation contradicts the team's own evidence — a process that ignores its own numbers cannot be trusted on any other number either
- D. It is irrelevant, because both models scored above 95%

23. The model decision:

"The regulator requires the bank to explain, in plain terms, why any particular application was rejected."

How does this requirement bear on the choice between the two models?

- A. It counts materially in favour of the tree-based model, which is far easier to explain and audit — though it can be answered for a neural network with additional explanation tooling, at a cost
- B. It rules out neural networks entirely, in banking and everywhere else
- C. It is a legal matter and has no bearing on which model is chosen
- D. It favours the neural network, since a more sophisticated model produces more convincing explanations

24. The model decision:

"A missed defaulter costs the bank roughly the outstanding principal; a wrongly rejected applicant costs the interest margin that customer would have generated."

What follows from this asymmetry?

- A. The threshold should be raised, so that as few customers as possible are rejected
- B. The rejection threshold should be lowered to catch more defaulters, accepting more false rejections, and the two models should be compared at that threshold rather than at the default 0.5
- C. The threshold should stay at 0.5, which weighs the two errors neutrally
- D. The asymmetry affects the business plan but not the choice of model

25. The model decision:

"The corrected evaluation arrives: Random Forest — 36% of defaulters caught, easy to explain, cheap to run. Neural Network — 37% caught, requires additional explanation tooling and more infrastructure."

What is the defensible call?

- A.** The Neural Network: 37% is higher than 36%, and higher is better
- B.** The Random Forest: a one-point difference is not a real advantage, and every other criterion points the same way
- C.** The Neural Network: it will improve as more data accumulates
- D.** Neither: both are below 50%, so no model should be deployed

When you have finished, check your answers against the version of this file that includes the answer key, and count your mistakes by part rather than in total: Part 1 tests how closely you have read the code, Part 2 tests whether plausible-sounding plans can mislead you, and Part 3 tests whether you set your criteria before reading the numbers.